Team Name : Project Hail

Team Leader Name : Poorna Sri Nandyala

Problem Statement : Problem Statement 15 – Forecasting and/or Nowcasting of Solar Flares using combined Soft and Hard X-ray data from Aditya-L1

Team Members

Team Leader:

Name : Poorna Sri Nandyala
College: Indian Institute of Information Technology, Sricity.

Team Member-1:

Name : Deekshitha Lingala
College: Indian Institute of Information Technology, Sricity.

Team Member-2:

Name : Rishita Mimmithi
College: Indian Institute of Information Technology, Sricity.

Team Member-3:

Name : Sriharsha Sripada
College: National Institute of Technology, Warangal.

Opportunity & Solution Alignment

Architectural Innovation & Scientific Merit:

- Direct Level-0 Nowcasting: Eliminates calibration latency by executing predictive models directly on raw Aditya-L1 orbital counts (SoLEXS & HEL1OS), bypassing traditional pre-processing bottlenecks.
- Physics-Informed Feature Space: Embeds astrophysical constraints into the model by extracting raw flare kinetics—specifically mapping flux velocity (v') and acceleration (v") to capture the Neupert Effect.
- Strict Chronological Validation: Employs rigorous, anti-leakage time-series splitting to simulate real-world operational environments, ensuring true predictive validity.

Operational Efficacy & Impact:

- High-Throughput Ingestion: Leverages DuckDB to process over 71 million rows of uncalibrated telemetry in sub-3 seconds on standard commercial hardware.
- Precision-Optimized ML: Utilizes an F1-score constrained architecture to suppress noise-floor false positives, achieving a robust True Skill Statistic (TSS > 0.82) up to 4 hours prior to peak flux

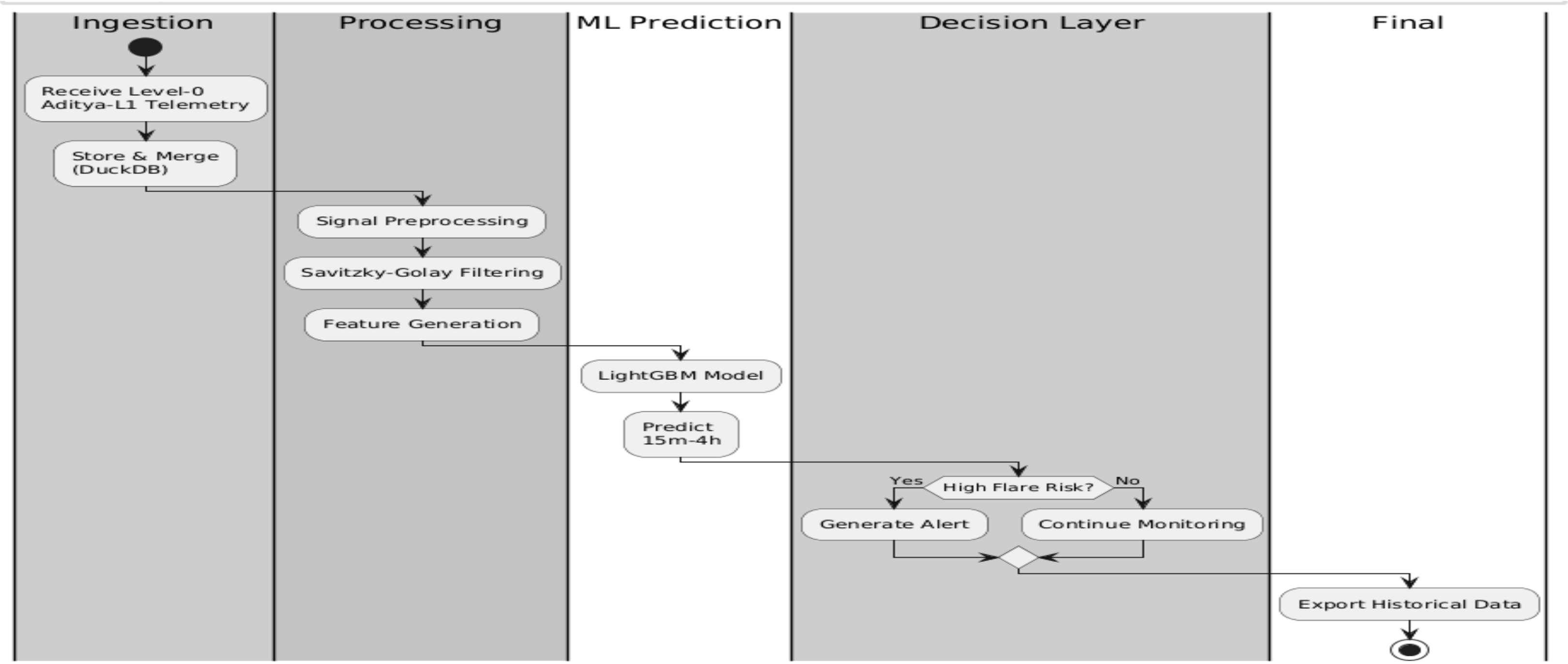
Operational Value:

- End-to-end pipeline from raw satellite telemetry → forecasting → real-time alerts & interactive dashboard.

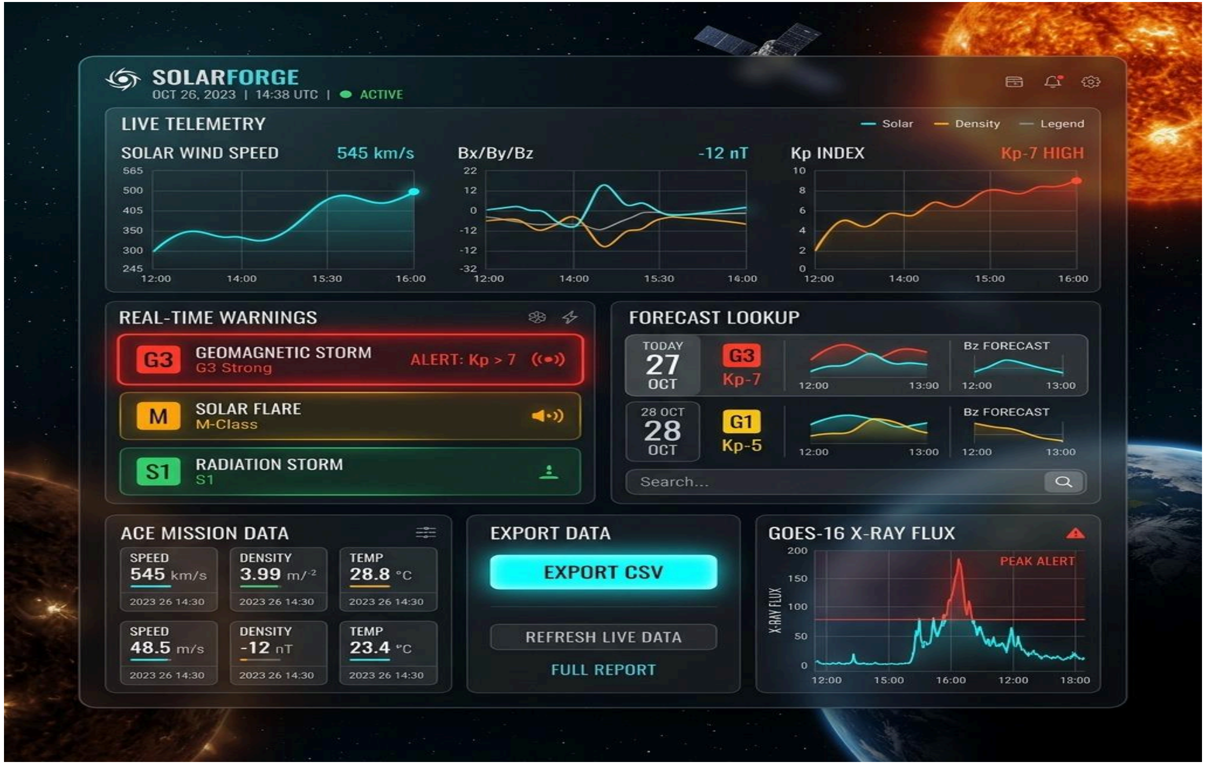
List of features offered by the solution

Multi-Temporal Forecasting Horizons • Multi-horizon prediction (15 min, 30 min, 1 hr, 2 hr, 4 hr). • Multi-class flare prediction (Nominal, C, M & X-Class). • Real-time probability estimation.	**Physics-Informed Analytics** • Savitzky–Golay signal filtering • Temporal derivative extraction • Velocity (v′) • Acceleration (v″) • Soft & Hard X-ray feature fusion
Mission Control Dashboard • Live telemetry visualization • Dynamic threat-level indicator • Interactive flare timeline • Event history • Audio & visual alert system	**Research & Deployment** • Telemetry export • Progressive Web App (PWA) • Offline support using Service Workers • Responsive desktop & mobile interface

Project flow Diagram



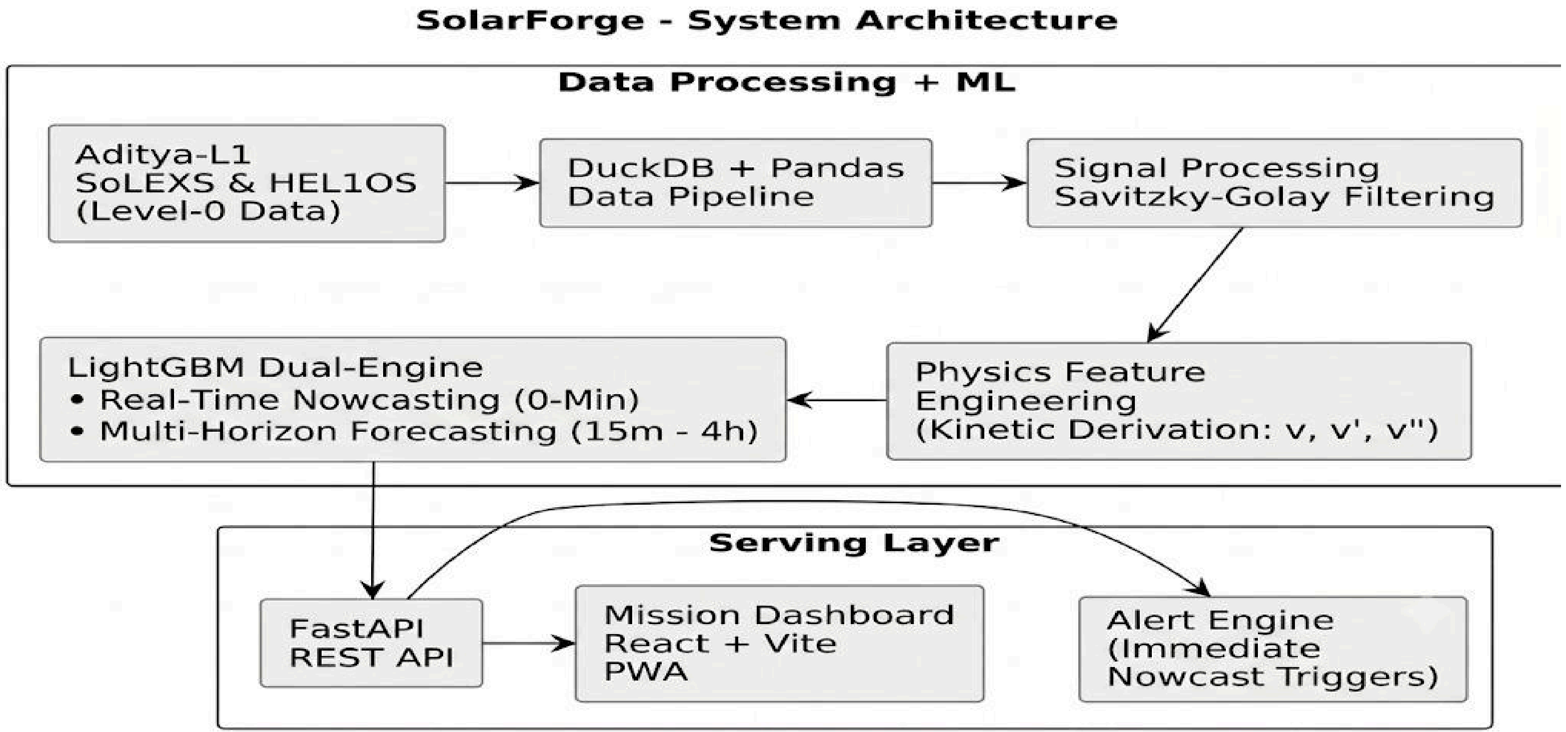
Wireframes/Mock diagrams of the proposed solution (optional)



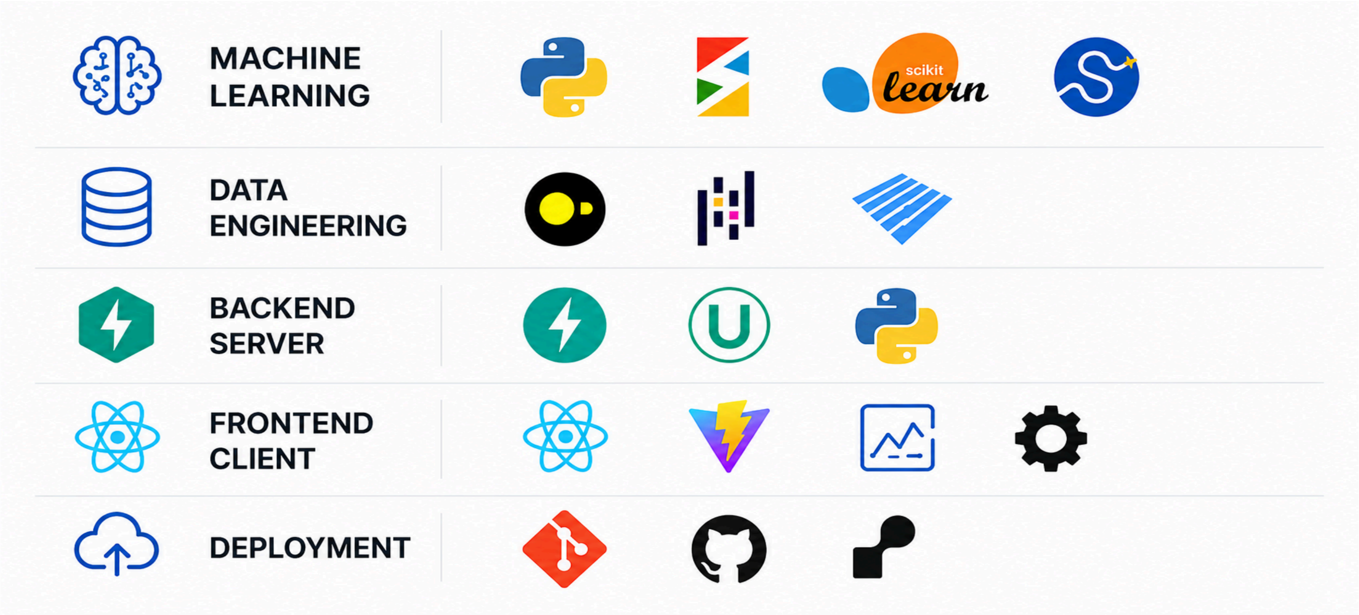
Key Modules

- Live Solar Telemetry
- AI Flare Prediction
- Alert System (Green → Red)
- X-ray Flux & Event Tracking
- Status
- Real-time system active
- AI integrated with dashboard
- End-to-end working prototype

Architecture diagram of the proposed solution



Technologies to be used in the solution:



Further Implementation:

Operational Lead Time: Measure alert-to-flare peak lead time for early warning.

Sensor Fusion: Integrate SUIT, VELC & X-ray data for better flare prediction.

Auto Calibration: Convert Level-0 telemetry into calibrated physical flux automatically.

Explainable AI: Live SHAP dashboard showing the key features behind each prediction.

Project Name : SolarForge

🔗Live Dashboard : https://solar-flare-prediction-bscp.onrender.com

🔗Project Codebase : https://github.com/SriHarsha25112006/Solar-Flare-Prediction

BHARATIYA ANTARIKSH HACKATHON

2026

THANK YOU